

```
In [2]: import pandas as pd
        from scipy import stats
        from statsmodels.stats import weightstats as stests
```

```
In [5]: data = pd.read_csv(r"E:\\pradnya's programming\\Domains\\DataAnalytics\\DA_clgAs
        data[['bp_before', 'bp_after']].describe()
        data.head(5)
```

```
Out[5]:
```

	patient	sex	agegrp	bp_before	bp_after
0	1	Male	30-45	143	153
1	2	Male	30-45	163	170
2	3	Male	30-45	153	168
3	4	Male	30-45	153	142
4	5	Male	30-45	146	141

```
In [6]: #T-test
        ttest,pval = stats.ttest_rel(data['bp_before'], data['bp_after'])
        print(pval)
```

```
0.0011297914644840823
```

```
In [7]: if pval<0.05:
        print("reject null hypothesis")
        else:
        print("accept null hypothesis")
```

```
reject null hypothesis
```

```
In [8]: #Chi-square Test
        data_chi = pd.read_csv(r"E:\\pradnya's programming\\Domains\\DataAnalytics\\DA_c
```

```
In [9]: contingency_table = pd.crosstab(data_chi["Gender"],data_chi["Like Shopping?"])
        print('contingency_table :-\n',contingency_table)
```

```
contingency_table :-
    Like Shopping?  No  Yes
Gender
Female            2    3
Male              2    2
```

```
In [10]: #Observed Values
        Observed_values = contingency_table.values
        print("Observed Values :-\n",Observed_values)
```

```
Observed Values :-
[[2 3]
 [2 2]]
```

```
In [11]: b=stats.chi2_contingency(contingency_table)
        print(b,"\n")
        Expected_values = b[3]
        print("Expected Values :-\n",Expected_values)
```

```
Chi2ContingencyResult(statistic=np.float64(0.0), pvalue=np.float64(1.0), dof=1, expected_freq=array([[2.22222222, 2.77777778],
[1.77777778, 2.22222222]]))
```

```
Expected Values :-
[[2.22222222 2.77777778]
 [1.77777778 2.22222222]]
```

```
In [12]: no_of_rows=len(contingency_table.iloc[0:2,0])
no_of_columns=len(contingency_table.iloc[0,0:2])
df11=(no_of_rows-1)*(no_of_columns-1)
print("Degree of Freedom:-",df11)
alpha = 0.05
```

Degree of Freedom:- 1

```
In [13]: from scipy.stats import chi2
chi_square=sum([(o-e)**2./e for o,e in zip(Observed_values,Expected_Values)])
chi_square_statistic=chi_square[0]+chi_square[1]
print("chi-square statistic:-",chi_square_statistic)
```

chi-square statistic:- 0.090000000000000008

```
In [14]: critical_value=chi2.ppf(q=1-alpha,df=df11)
print('critical_value:',critical_value)
```

critical_value: 3.841458820694124

```
In [15]: #p-value
p_value=1-chi2.cdf(x=chi_square_statistic,df=df11)
print('p-value:',p_value)
```

p-value: 0.7641771556220945

```
In [16]: print('Significance level: ',alpha)
print('Degree of Freedom: ',df11)
print('chi-square statistic:',chi_square_statistic)
print('critical_value:',critical_value)
print('p-value:',p_value)
```

Significance level: 0.05
Degree of Freedom: 1
chi-square statistic: 0.090000000000000008
critical_value: 3.841458820694124
p-value: 0.7641771556220945

```
In [17]: if chi_square_statistic>=critical_value:
    print("Reject H0,There is a relationship between 2 categorical variables")
else:
    print("Retain H0,There is no relationship between 2 categorical variables")

if p_value<=alpha:
    print("Reject H0,There is a relationship between 2 categorical variables")
else:
    print("Retain H0,There is no relationship between 2 categorical variables")
```

Retain H0,There is no relationship between 2 categorical variables
Retain H0,There is no relationship between 2 categorical variables

In []: